SHIVAJI UNIVERSITY,KOLHAPUR

NAAC A++ Grade with CGPA 3.52

Multiple Entry and Multiple Exit Option(NEP-2020)

Syllabus for

B.Sc. Artificial Intelligence & Machine Learning

(Apprenticeship Embedded Degree Programmes)

(Under Faculty of Science and Technology)

PART- I SEMESTER-I & II

(Syllabus to be implemented from Academic year 2025-26)

B.Sc. Artificial Intelligence & Machine Learning (AEDP)

Academic year 2025-26

1. Title: B.Sc. Artificial Intelligence & Machine Learning (AEDP)
2. Year of implementation: Syllabus will be implemented from June 2025 onwards
3. Duration: B.Sc. Artificial Intelligence & Machine Learning (AEDP)

Part- I. The duration of course shall be one year (Two semesters).

4. Pattern: Pattern of examination will be semester
5. Structure Of Course:

(Syllabus to be implemented from Academic year 2025-26)

Shivaji University, Kolhapur

B.Sc.-M.Sc. Artificial Intelligence & Machine Learning (AI & ML) (AEDP)

(Under Faculty of Science and Technology)

Program Outcomes

Upon successful completion of the B.Sc.-M.Sc. Artificial Intelligence & Machine Learning (AI & ML), the student should have met the following Outcomes:

1. Apply skills and concepts of Computer Science to understand, design and code computing models from problem statements.
2. Ability to design, develop and implement real world problems to software solutions
3. Apply mathematical and statistical models to explore and analyze the massive amount of data in various domains.
4. Design and develop Artificial Intelligence and Machine Learning (AIML) systems for analysing the varied, voluminous data in various domains.
5. Recognize the importance of innovation and develop a critical and research thinking approach to develop leading innovative intelligent software products.
6. Recognize social, professional, cultural, humane and ethical values and issues involved in the use of software technology and consider these adequately in developing and deploying systems
7. Be self-motivated and enhance self-learning as well as capability to adapt to emerging technologies and create innovative solutions for challenging issues of society.

Program Specific Outcome (PSO)

1. An ability to enhance the application of knowledge of theory subjects in diverse fields.
2. Encouraging students to convert their start-up idea to reality by implementing
3. Focuses on preparing the student for roles in computer applications and the IT industry.
4. Develop programming skills, networking skills, learn applications, packages, programming languages, and modern techniques of IT.

5. Information about various computer applications and the latest development in IT and communication systems is also provided.
6. Ability to identify, formulates, analyze and solve problems of programming using different languages.
7. Take up self-employment in the Indian & global software market.
8. To pursue a career in the corporate sector can opt for M.Sc, MCA.
9. The student will be able to know various issues, and the latest trends in technology development and thereby innovate new ideas and solutions to existing problems.

Programme Educational Objectives (PEOs)

1. Develop intelligent software's solutions to handle uncertainty.
2. Promote research in intelligent technology and concepts.
3. Participate in life-long learning for effective professional growth and demonstrate leadership qualities.

1. Introduction

The first step towards a graduate professional degree in flourishing area of Artificial Intelligence and Machine Learning. The B.Sc.-M.Sc. Artificial Intelligence & Machine Learning (AI & ML) (Five years Integrated Programme) will provide students with the Knowledge, Skills, and Abilities to meet the demands of the ever-changing IT industry. Beginning with the fundamental concepts of Computing, the programme enables students to appreciate the application of AI and ML techniques in Practise through fascinating modes of learning. B.Sc.-M.Sc. Artificial Intelligence & Machine Learning (AI & ML) (Five years Integrated Programme) will empower students to become future technology leaders who will truly transformative and embrace rapid technology growth and lead innovation, whether it be within the organization as a new start-ups or outside the organization's boundaries as an IT consultant.

2. Multiple Entry and Multiple Exit Option (NEP-2020):

- After successful completion of First year of M.Sc. AI and ML (Five years Integrated Programme) Student can exit with under graduate certificate in Computer Science.
- After successful completion of Second year of MSc. AI and ML (Five years Integrated Programme) Student can exit with under graduate Diploma in Computer Science.

- After successful completion of Third year of M.Sc. AI and ML (Five years Integrated Programme) Student can exit with B.Sc. AI & ML degree programme.
- After successful completion of fourth year of M.Sc. AI and ML (Five years Integrated Programme) Student can exit with Post Graduate Diploma in AI & ML.
- After successful completion of fifth year of M.Sc. AI & ML (Five years Integrated Programme) Student will be awarded M.Sc. AI & ML degree.

3. Medium of Instruction:

The medium of instruction will be English only

4. Admission Procedure

To be eligible for admission to the Master of Science AI and ML (Five years Integrated Program) a candidate must have:

- ✓ Passed HSC (10+2) from Science stream.

  or
- ✓ Diploma (10+3) in any stream

  Reservation policy will be as per university norms.

5. Course Structure:

Lectures and Practical should be conducted as per the scheme of lectures and practicals indicated in the course structure.

6. Teaching and Practical Scheme

a) Each contact session for teaching 60 minutes each.

b) One Practical Batch should be of 30 students.

c) Practical evaluation should be conducted after the commencement of University examination.

7. Project Work:

a) Project work may be done individually or in groups in case of bigger projects. However if project is done in groups, each student must be given a responsibility for a distinct module and care should be taken to see the progress of individual modules is independent of others.

b) Students should take guidance from assigned guide and prepare a Project Report on "Project Work" in two copies to be submitted to the Head of the Department.

c) The project report should contain an Introduction to Project, which should clearly explain the project scope in detail. Also DFDs, ERDs, UML diagrams, database designs and a list of output reports should be included.

d) The project Work should be of such a nature that it could prove useful or should be relevant from the commercial/Societal angle.

e) The project report will be duly accessed by the assigned guide and internal marks will be communicated by the Director of the Institute/Head of the Department.

f) The project report should be prepared in a format prescribed by the University, which also specifies the contents and methods of presentation. IEEE Computer Society templates are recommended in this regard.

g) The external viva shall be conducted by a panel of minimum two examiners out of which one will be external and other will be internal examiner.

8. Assessment

1. The project will be evaluated by the university appointed examiners both internal as well as external.
2. The final practical examination will be conducted by the university appointed examiners both internal as well as external at the end of semester for each lab course and marks will be submitted to the university by the panel.
3. The practical examination will be conducted semester wise in order to maintain the relevance of the respective theory course with laboratory course.
4. The final examinations shall be conducted at the end of the semester.
5. Nature of question paper:

   Nature of question paper is as follows for University end semester examination

❖ **Theory Examination:**

Year	**Semester**	**Activity**	**Marks (2 Credits)**
1	I & II	1. Home Assignment 2. Class Assignment (Tutorial type) 3. Quiz 4. Midterm Test*	5 5 5 5
2	III & IV	1. Oral Examination 2. Group Discussion 3. Seminar 4. Midterm Test*	5 5 5 5

3	V & VI	1. Case Study 2. Field Work 3. Book Review/ Poster Presentation 4. Midterm Test*	5 5 5 5
4	VII & VIII	1. Seminar 2. Case Study/ Problem Solving 3. Book Review/ Poster Presentation 4. Midterm Test*	5 5 5 5
Note: 1. Midterm Test shall be conducted after completion of each unit 2. Book Review (Only from Reference Book)			

Each paper will carry 30 Marks

1. Question No.1 is compulsory and is of multiple choice questions. There will be 6 multiple choice questions each carrying 1 mark
2. Question No.2 will have 3 questions out of which 2 questions need to be solved. Each carries 6 Marks
3. Question No.3 will have 6 questions out of which 4 questions need to be solved. Each carries 3 Marks

❖ **Practical Examination:**

1. Each paper carries 30 Marks
2. Duration of Practical Examination: 2 Hrs
3. Nature of Question paper: There will be 3 questions out of which any 2 questions to be attempted and each question carries 15 Marks.
4. Internal Marks: 20(Journal :10 Marks, Internal Viva:10 Marks)
5. Practical evaluation should be conducted after the commencement of University examination by External examination.
6. Theory/Practical Exams of Open Elective will be conducted by Internal Evaluator.

9. Standard of Passing:

i) Minimum 35% marks in each subject. There shall be separate passing for theory and practical.

ii) Admission to B.Sc.-M.Sc. Artificial Intelligence & Machine Learning (AI & ML) Part II is allowed even if the students fails in 20 subjects of First year B.Sc.-M.Sc. Artificial Intelligence & Machine Learning (AI & ML)

iii) Admission to B.Sc.-M.Sc. Artificial Intelligence & Machine Learning (AI & ML) part III is allowed even if the students fail in 20 subjects of B.Sc.-M.Sc. Artificial Intelligence & Machine Learning (AI & ML) Part II. But no student is allowed to take admission to third year of B.Sc.-M.Sc. Artificial Intelligence & Machine Learning (AI & ML) unless they clears all the papers of first year.

10. Board of Paper Setters /Examiners:

For each Semester end examination there will be a board of Paper setters and examiners for every course. While appointing paper setter /examiners, care should be taken to see that there is at least one person specialized in each unit of the course.

11. Credit system implementation:

As per the University norms

12. Clarification of Syllabus:

The syllabus committee should meet at least once in a year to study and clarify any difficulties from the Institutes.

13. Eligibility of Faculty:

MCA (from any faculty) or M.Sc. (Computer Science) or M.Tech.(CS) with at least B+ or equivalent

14. Revision of Syllabus:

As the computer technology experience rapid rate of obsolescence of knowledge, revision of the syllabus should be considered every two/three years.

15. Award of Class:

There will be numerical marking on each question. At the time of declaration of the result the marks obtained by the candidate is converted into grade point as shown below;

Grading: Shivaji University has introduced a Seven-point grading system as follows

Sr. No.	Marks Range out of 50	Grade Point	CGPA	Letter grade
1.	43-50	10	9.50-10.00	O: Outstanding
2.	38-42	9	8.50-9.49	A+:Excellent
3.	33-37	8	7.50-8.49	A:Very Good
4.	28-32	7	6.50-7.49	B+:Good
5.	23-27	6	5.50-6.49	B: Average
6.	18-22	5	4.50-5.49	C:Satisfactory
7.	0-17	0	0.0-4.49	F:Fail
8.	Absent	0	------	-----

B.Sc.-M.Sc. Artificial Intelligence & Machine Learning (AI & ML) (Five years Integrated Programme)

Academic year 2025-26

Syllabus to be implemented from Academic Year 2025-26

1. Title: B.Sc.-M.Sc. Artificial Intelligence & Machine Learning (AI & ML)
2. Year of implementation: Syllabus will be implemented from June 2023 onwards
3. Duration: B.Sc.-M.Sc. Artificial Intelligence & Machine Learning (AI & ML)

   Part- I. The duration of course shall be one year (Two semesters).
4. Pattern: Pattern of examination will be semester
5. Structure Of Course:

Multiple Entry and Multiple Exit Option (NEP-2020)

B.Sc. Artificial Intelligence & Machine Learning (AEDP)

Programme Structure

B.Sc. Part - I (Level-4.5)

Part – I Semester – I

SEMESTER-I (Duration- Six Month)								
Sr. No.	Course Code	**Teaching Scheme**			**Examination Scheme**			
		Theory and Practical			University Assessment (UA)		Internal Assessment (IA)	
		Lectures (Per week)	Hours (Per week)	Credit	Maximum Marks	Minimum Marks	Maximum Marks	MinimumMarks
1	Subject I DSC I: Programming Paradigm I	2	-	2	30	11	20	07
2	Subject I DSC II: Fundamentals of Operating System	2	-	2	30	11	20	07
3	Subject I Practical I: Practical based on Subject I DSC I	-	4	2	30	11	20	07
4	Subject II DSC I: Introduction to Statistics	2	-	2	30	11	20	07
5	Subject II DSC II: Probability Distribution	2	-	2	30	11	20	07
6	Subject II Practical I: Practical based on Subject II DSC I & Subject II DSC II	-	4	2	30	11	20	07
7	Subject III DSC I: Fundamentals of Digital Electronics	2	-	2	30	11	20	07
8	Subject III DSC II: Data Communication	2	-	2	30	11	20	07
9	Subject III Practical I: Practical based on Subject III DSC I & Subject III DSC II	-	4	2	30	11	20	07
10	OE I: Office Automation-I Lab	-	4	2	30	11	20	07
11	IKS I: Vedic Mathematics	2	-	2	30	11	20	07
	Total (A)			22	330		220	

SEMESTER-II (Duration- Six Month)								
Sr. No.	Course Code	Teaching Scheme			Examination Scheme			
		Theory and Practical			University Assessment (UA)		Internal Assessment (IA)	
		Lectures (Per week)	Hours (Per week)	Credit	Maximum Marks	Minimum Marks	Maximum Marks	Minimum Marks
1	Subject I DSC III: Programming Paradigm II	2	-	2	30	11	20	07
2	Subject I DSC IV: System Analysis and Design	2	-	2	30	11	20	07
3	Subject I Practical II: Programming Paradigm II Lab	-	4	2	30	11	20	07
4	Subject II DSC III: Statistical Analysis	2	-	2	30	11	20	07
5	Subject II DSC IV: Advanced Excel	2	-	2	30	11	20	07
6	Subject II Practical II: Practical based on Subject II DSC III & Subject II DSC IV	-	4	2	30	11	20	07
7	Subject III DSC III: Computer Architecture	2	-	2	30	11	20	07
8	Subject III DSC IV: Computer Network	2	-	2	30	11	20	07
9	Subject III Practical II: Practical based on Subject III DSC III and Subject III DSC IV	-	4	2	30	11	20	07
10	OE II : Basics of HTML	-	4	2	30	11	20	07
11	VEC I: Democracy, Election and Constitution	2	-	2	30	11	20	07
	Total (B)			22	330		220	
	Total (A+B)			**44**	**660**		**440**	

Multiple Entry and Multiple Exit Option (NEP-2020)
B.Sc. AI & ML (AEDP) Programme Structure
B.Sc. AI & ML (AEDP) Part - II (Level-5.0)

SEMESTER-III (Duration- Six Month)								
Sr. No.	Course Code	**Teaching Scheme**			**Examination Scheme**			
		Theory and Practical			University Assessment (UA)		Internal Assessment (IA)	
		Lectures (Per week)	Hours (Per week)	Credit	Maximum Marks	Minimum Marks	Maximum Marks	Minimum Marks
1	Subject I Major V Data Structure	2	-	2	30	11	20	07
2	Subject I Major VI Database Management System	2	-	2	30	11	20	07
3	Subject I Practical III Practical Based on Subject I Major V & Major VI	-	4	2	30	11	20	07
4	Subject II Minor V Exploratory Data Analysis	2	-	2	30	11	20	07
5	Subject II Minor VI Advanced Python Programming	2	-	2	30	11	20	07
6	Subject II Practical III Practical Based on Subject Minor VI	-	4	2	30	11	20	07
7	OE – III (T/P) Discrete Mathematics-I	2	-	2	30	11	20	07
8	VSC – I (P) Major specific Web Design using JavaScript	-	4	2	30	11	20	07
9	SEC-I (T/P): Core Java Programming	-	4	2	30	11	20	07
10	AEC-I Verbal Communication	2	-	2	30	11	20	07
11	CC-I Yoga	2	-	2	30	11	20	07
	Total (A)			22	330		220	

SEMESTER-IV (Duration- Six Month)								
Sr. No.	Course Code	Teaching Scheme			Examination Scheme			
		Theory and Practical			University Assessment (UA)		Internal Assessment (IA)	
		Lectures (Per week)	Hours (Per week)	Credit	Maximum Marks	Minimum Marks	Maximum Marks	Minimum Marks
1	Subject I Major VII Advanced Data Structures	2	-	2	30	11	20	07
2	Subject I Major VIII Relational Database Management System	2	-	2	30	11	20	07
3	Subject I Practical IV Practical Based on Subject I Major VII & Major VIII	-	4	2	30	11	20	07
4	Subject II Minor VII Feature Engineering	2	-	2	30	11	20	07
5	Subject II minor VIII Basics of Artificial Intelligence	2	-	2	30	11	20	07
6	Subject II Practical IV Practical Based on Subject II Minor VII	-	4	2	30	11	20	07
7	OE – III (T/P) Discrete Mathematics-II	2	-	2	30	11	20	07
8	SEC-II(T/P) Advanced Java Programming	-	4	2	30	11	20	07
9	AEC-II Business Communication	2	-	2	30	11	20	07
10	VEC-II Environmental Studies	-	4	2	30	11	20	07
11	CEP-I NSS/NCC	2	-	2	30	11	20	07
	Total (A)			22	330		220	

Multiple Entry and Multiple Exit Option (NEP-2020)

B.Sc. AI & ML (AEDP) Programme Structure

B.Sc. AI & ML (AEDP) Part - III (Level-5.5)

SEMESTER-V (Duration- Six Month)								
Sr. No.	Course Code	**Teaching Scheme**			**Examination Scheme**			
		Theory and Practical			University Assessment (UA)		Internal Assessment (IA)	
		Lectures (Per week)	Hours (Per week)	Credit	Maximum Marks	Minimum Marks	Maximum Marks	Minimum Marks
1	Subject I Major IX Artificial Intelligence	2	-	2	30	11	20	07
2	Subject I Major X Fundamentals of Machine Learning	2	-	2	30	11	20	07
3	Subject I Practical V Practical Based on Subject I Major X	-	4	2	30	11	20	07
4	Subject II Major I Elective ✓ Web Application Development ✓ PHP	2	-	2	30	11	20	07
5	Subject II Major Practical-1 Elective Practical Based on Subject II Major I Elective	-	4	2	30	11	20	07
6	OE – V Numerical Methods	2	-	2	30	11	20	07
7	VSC-II Major specific Front End Development	2	-	2	30	11	20	07
8	AEC-III Corporate Culture & Communication	2	-	2	30	11	20	07
9	OJT On Job Training	-	4	2	30	11	20	07
	Total (A)			22	330		110	

SEMESTER-VI (Duration- Six Month)									
Sr. No.	Course Code	Teaching Scheme			Examination Scheme				
		Theory and Practical			University Assessment (UA)			Internal Assessment (IA)	
		Lectures (Per week)	Hours (Per week)	Credit	Maximum Marks	Minimum Marks	Exam minutes	Maximum Marks	Minimum Marks
1	Apprenticeship/ Internship	-	-	22	400	140	-	150	53
	Total (A)			22	400			150	

B.Sc.-M.Sc. Artificial Intelligence & Machine Learning (AI & ML)
Multiple Entry and Multiple Exit Option
(NEP-2020)
PART I SEM I

Title of course: Programming Paradigm I

Course Outcomes:

After completion of this course students will be able to;

1. Understand principles of Python.
2. Understand installation of Python.
3. Explore concepts and modules in Python.
4. Implement error handling using try-except for robust coding.
5. Apply control structures like if-else and loops in Python programs.
6. Perform file operations including reading and writing data.

Unit I: **(15 HOURS)**

Introduction – Installing Python, Features and principles of Python, applications of Python, Simple program using Python, Expressions and Values, Variables and Computer Memory, error detection, Multiple line statements, Identifiers ,Keywords , Statements and Expressions – Variables Operators , Arithmetic operators, Assignment operators , Comparison operators – Logical operators ,Bitwise operators - Precedence and Associativity, Data types , Number, Booleans , Strings , Indentation, Comments , Single line comment , Multiline comments - Reading Input , Print output ,Type Conversions.

UNIT II: **(15 HOURS)**

Control flow statements If statement , If else statement , nested if statement - while loop , for loop ,continue and break statements ,catching exceptions using try and except statement ,syntax errors , exceptions, exception handling ,Strings function. Nesting loops in loops, looping until a condition is reached, and repetition based on user input, controlling Loops using Break and Continue. Reading and Writing files: kinds of files, Opening a File, techniques for reading files, files over the internet, Writing files, file reading techniques.

Reference Books:

1. Learning Python, Fourth Edition, Mark Lutz, O'Reilly publication.
2. Introduction to Python for Computational Science and Engineering (A beginner's guide), Hans Fangohr.
3. John V Guttag. "Introduction to Computation and Programming Using Python", Prentice Hall of India.
4. R. Nageswara Rao, "Core Python Programming", Dreamtech

B.Sc.-M.Sc. Artificial Intelligence & Machine Learning (AI & ML)
Multiple Entry and Multiple Exit Option
(NEP-2020)
PART I SEM I

Title of course: Fundamentals of Operating System

Course Outcomes:

After completion of this course students will be able to;

1. Understand fundamental operating system.
2. Learn types of operating systems, Scheduling.
3. Learn Process Synchronization, Deadlock Prevention, and Avoidance.
4. Understand multithreading and models for multicore programming.
5. Apply synchronization techniques using semaphores, mutexes, and monitors.
6. Evaluate scheduling algorithms and deadlock handling methods.

UNIT I **(15 HOURS)**

Evolution of Operating Systems: Introduction, Types of operating systems, Computer System Organization, Architecture, Structure, Operations, Process Management, Memory Management, Storage Management, Kernel Data Structures, Computing Environments. Operating System. Structures: Services, System Calls, Types, Operating System Structure, System Boot. Processes: Process Concept, Scheduling, Operations, Interprocess Communication. Multithreaded Programming: Multicore Programming, Multithreading Models.

UNIT II **(15 HOURS)**

Process Synchronization: The Critical-Section Problem, Peterson's Solution, Synchronisation Hardware, Mutex Locks, Semaphores, Classic Problems of Synchronization, Monitors, Synchronization Examples. Process Scheduling: Criteria, Scheduling Algorithms, Multi-Processor Scheduling, Real-time CPU Scheduling. Deadlocks: System model, Characterization, Methods for handling deadlocks, Deadlock Prevention, Avoidance, Detection and Recovery from deadlock.

Reference Books:

1. William Stallings, "Operating Systems-Internals and Design Principles", Pearson, IX Edition, 2018
2. D M Dhamdhere: Operating Systems – A Concept Based Approach, III Edition, Tata McGraw– Hill, 2015.
3. Harvey M Deitel, Paul J Deitel, Dr Choffnes, Operating Systems, Pearson Education

B.Sc.-M.Sc. Artificial Intelligence & Machine Learning (AI & ML)
Multiple Entry and Multiple Exit Option
(NEP-2020)
PART I SEM I

Title of course: Practical based on Subject I DSC I

Course Outcomes:

After completion of this course students will be able to;

1. To make the student learn a programming language.
2. To learn problem solving techniques.
3. To teach the student to write programs in Python and to solve the problems
4. Implement control flow using conditional statements and loops.
5. Apply string functions and nested loops to solve programming tasks.
6. Read from and write to files using standard file handling techniques in Python.

Lab work is based on Python Programming Language. This laboratory course will consist of 25 to 30. Programming exercises with focus on covering the hands-on aspects covered in theory course.

B.Sc.-M.Sc. Artificial Intelligence & Machine Learning (AI & ML)
Multiple Entry and Multiple Exit Option
(NEP-2020)
PART I SEM I

Title of course: Introduction to Statistics

Course Outcomes:

After completion of this course students will be able to;

1. Acquire knowledge of Meaning and Scope of Statistics, various statistical organizations.
2. Understand the basic knowledge of data collection and various statistical elementary tools.
3. Understand the basic concept of Measures of Central Tendency and Dispersion.
4. Organize and represent data using frequency distributions and classification methods.
5. Represent statistical data graphically using histograms, polygons, ogives, and pie charts.
6. Calculate measures of dispersion and skewness using appropriate statistical methods.

UNIT I **(15 HOURS)**

Introduction to Statistics: Concept, Definition of Statistics and Scope of statistics, Nature of Data: Primary and Secondary data, Qualitative and quantitative data, Discrete and continuous data, frequency, cumulative frequency, frequency distribution..Population and Sample: Statistical population. Finite population, Infinite population, Census method, Sample, sampling method, Advantages of sampling method over census method. SRS, SRSWR and SRSWOR Representation of Data: Discrete frequency distribution, Continuous frequency distribution, Cumulative frequency distribution, Inclusive and Exclusive methods of classification, Open end classes, illustrative examples. Representation of data by graphical method: Histogram, frequency polygon, frequency curve, Ogive curve. Representation of data by diagrammatic Method: Bar diagram (Simple), Pie chart.

UNIT II **(15 HOURS)**

Measures of central Tendency: Meaning of averages, Requirements of good average. Arithmetic mean (A.M.), Combined mean, Median, Quartiles, Mode, Relation between mean, median and mode. Merits and Demerits of Mean, Median and Mode, determination of Median and Mode by Graph. Numerical Examples.Measures of Dispersion: Meaning of dispersion, Absolute and Relative measures of dispersion Requirements of good dispersion. Q.D, M.D, S.D. Variance and Combined variance, Coefficient of Variation (C.V) Concept of Skewness, Numerical Examples

Reference Books:

1. Statistical Methods, by Dr. S. P. Gupta, Sultan Chand and Sons Publication.
2. Introduction to Statistics, by C.B. Gupta.
3. Mathematical Statistics, by H.C. Saxena and J.N. Kapur.
4. Business Statistics, by S.S. Desai.
5. Business Statistics, by G.V. Kumbhojkar.
6. Fundamentals of Statistics, by S.C.Gupta.
7. Business Statistics-SIM- Shivaji University, Kolhapur

B.Sc.-M.Sc. Artificial Intelligence & Machine Learning (AI & ML)
Multiple Entry and Multiple Exit Option
(NEP-2020)
PART I SEM I

Title of course: Probability Distribution

Course Outcomes:

1. acquire knowledge of concepts of probability.
2. understand concept of random variable and its probability distributions.
3. Apply classical, addition, multiplication, and conditional probability rules.
4. Understand and interpret probability mass functions (p.m.f.) and density functions.
5. Analyze binomial distribution and compute its mean, variance, and probabilities.
6. Evaluate Poisson distribution and apply its properties to real-life scenarios.

UNIT I **(15 HOURS)**

Probability: - Trial, Sample Space, Events, Types of event, Classical definition of Probability, Addition law of probability, Conditional Probability. Multiplication law of probability, Concept of independent events. Examples without use of permutations and computations. Concept of Probability Mass Function (p.m.f) and Probability Density Function (p.d.f), Mean and variance. Simple, examples.

UNIT II **(15 HOURS)**

Concept of Random variable and Probability distribution. Binomial Distribution, Definition, mean, variance, additive property of binomial distribution and examples. Poisson Distribution, Definition, mean, variance, additive property of Poisson distribution and examples. Simple examples to find probabilities and parameters. Normal Distribution Definition, mean, variance and Limiting case (binomial and Poisson distribution)Definition of Standard Normal variate and its p.d.f. Properties of normal curve, Examples to find probabilities for given area under standard normal curve.

Reference Books:-

1) Elements of Statistics by D. N. Elance.

2) Introduction to Statistics, by C.B. Gupta.

3) Mathematical Statistics, by H.C. Saxena and J.N. Kapur.

B.Sc.-M.Sc. Artificial Intelligence & Machine Learning (AI & ML)
Multiple Entry and Multiple Exit Option
(NEP-2020)
PART I SEM I

Title of course: Practical based on Subject II DSC I & Subject II DSC II

Course Outcomes:

After completion of this course students will be able to;

1. Study measures of Central Tendency and Dispersion.
2. Apply binomial and Poisson distributions.
3. Solve problems using classical, addition, multiplication, and conditional probability rules.
4. Apply binomial distribution to calculate probabilities, mean, and variance.
5. Calculate measures of central tendency such as mean, median, quartiles, and mode using sample data.
6. Determine and compare absolute and relative measures of dispersion including Q.D., M.D., S.D., and variance.

Lab work is based on Introduction to Statistics & Probability Distribution. This laboratory course will consist of 25 to 30. Programming exercises with focus on covering the hands-on aspects covered in theory course.

B.Sc.-M.Sc. Artificial Intelligence & Machine Learning (AI & ML)
Multiple Entry and Multiple Exit Option
(NEP-2020)
PART I SEM I

Title of course: Fundamentals of Digital Electronics

Course Outcomes:

After completion of this course students will be able to;

1. Understand the concept of number system.
2. Understand the functionality of logic gates.
3. Describe binary codes such as Gray, Excess-3, ASCII, EBCDIC and their applications in data representation.
4. Analyze binary number representations using signed/unsigned formats and complements (1's, 2's, 9's).
5. Understand the concept of error detection and correction using Hamming code.
6. Analyze logic families (TTL, CMOS), their characteristics, and applications in digital systems.

UNIT I **(15 HOURS)**

Number Systems: Introduction to Decimal, Binary, Hexadecimal Number systemInterconversion from one system to Another (examples),code, Gray code, Exess-3 co de, ASCII code, EBCDIC code ,Concept of parity bit ,Signed and unsigned numbers represe ntation ,1's & 2's complement of binary numbers, 9's complement and binary arithmetic, Hamming code for error correction & detection

UNIT II **(15 HOURS)**

Logic Gates:AND, OR, NOT, NOR, NAND, EX-OR (Symbol, Expression and Truth Table), Application of EX-OR gate , Boolean algebra and identities ,De Morgan's theorem and Inter conversion of logic Gates (NAND and NOR), Simplifications of logic expressions using : Boolean algebra, K map with examples, Introduction to logic families (TTL, ECL, CMOS), TTL NAND gate & CMOS NOT gate, Input output parameters Logic levels, switching speed, propagation delay, power dissipation, noise margins and fan in-out of TTL and CMOS, Tristate logic (inverter & buffer)

Reference Books:

1. Digital Fundamentals by Morris and Mano, PHI Publication

2. Fundamental of digital circuits by A.Anandkumar,PHI Publication

3. Digital Fundamentals by Floyd & Jain, Pearsons Pub

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(NEP-2020)
PART I SEM I

Title of course: Data Communication

Course Outcomes:

After completion of this course students will be able to;

1. Understand the various components of a computer network and its functionality.
2. Become familiar with layered communication architectures (OSI and TCP/IP).
3. Familiar with network basics concepts like protocols, topology etc.
4. Understand the OSI reference model and the functionality of each layer.
5. Illustrate token passing techniques and the working of FDDI networks.
6. Compare the OSI and TCP/IP models and explain internetworking concepts.

UNIT I **(15 HOURS)**

Introduction, Network topologies, network classifications, Layered network architecture, LAN, WAN, MAN, The telephone network fundamental of communication theory, Data transmission modes, Network topologies, Transmission Media, Guided media, twisted-pair cable, coaxial cable, fiber-optic cable. Unguided media (wireless), radio waves, microwaves, infrared, Asynchronous and synchronous transmission

UNIT II **(15 HOURS)**

Overview of OSI reference model, it's all layer's services. Token passing – Token ring, Token bus, Token passing, (priority systems). Fiber Distributed Data Interface (FDDI).Overview of TCP/IP, Introduction to TCP/IP and internetworking, operations related protocols and sockets,Connection-oriented and connectionless Services, service primitives. OSI protocols, TCP/IP protocols. Physical Layer: Physical Layer Basic Concepts - Bit rate, bit length, base band transmission, Switching Circuit switching, Packet Switching, Message switching.

Reference Books:

1. Black C "Computer networks protocols, standards and Interface", prentice hall of India, 1996
2. Stlling W, "Computer communication network" (4th Edition), prentice hall of India, 1993
3. Tanenbaum A.S. "Computer Network", prentice hall of India, 1981
4. Forouzan, "TCP/IP Protocol Suite", Tata McGraw Hill.

B.Sc.-M.Sc. Artificial Intelligence & Machine Learning (AI & ML)
Multiple Entry and Multiple Exit Option
(NEP-2020)
PART I SEM I

Title of course: Practical based on Subject III DSC I & Subject III DSC II

Course Outcomes:

After completion of this course students will be able to;

1. The students will be able to perform number system conversions.
2. The students will understand various types of basic, combinational & universal logic gates
3. To design and analysis of the routing Protocols.
4. To understand key Internet applications and their protocols.
5. Demonstrate the configuration and working of LAN, WAN, and MAN networks using basic tools.
6. Analyze connection-oriented and connectionless communication using TCP/UDP protocols.

Lab work is based on Fundamentals of digital Electronics & Data Communication. This laboratory course will consist of 25 to 30. Programming exercises with focus on covering the hands-on aspects covered in theory course.

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Multiple Entry and Multiple Exit Option
(NEP-2020)
PART I SEM I

Title of course: OE-I (P) Office Automation-I

Course Outcomes:
After completion of this course students will be able to;

1. Understand the basics of MS Word.
2. Understand the working of PowerPoint.
3. Apply formatting styles, slide layouts, and multimedia elements in PowerPoint presentations.
4. Demonstrate the use of tables, symbols, and clip arts in MS Word documents.
5. Apply formatting styles, slide layouts, and multimedia elements in PowerPoint presentations.
6. Identify and describe the use of animation and transition effects to enhance presentations.

UNIT I **(15 HOURS)**

MS Word: Working with Documents -Opening & Saving files, Editing text documents, Formatting page & setting Margins, Formatting Documents - Setting Font styles, Font selection- style, size, colour etc, Type face - Bold, Italic, Underline, Case settings, Highlighting, Special symbols, Setting Paragraph style, Alignments, Indents, Line Space, Margins, Bullets & Numbering. Setting Page style Creating Tables- Table settings, Borders, Alignments, Insertion, deletion, Merging, Splitting, Sorting, and Formula, Drawing - Inserting Clip Arts, Pictures/Files etc.

UNIT II **(15 HOURS)**

MS Power point: Introduction to presentation – Opening new presentation, Different presentation templates, setting backgrounds, selecting presentation layouts. Creating a presentation, Setting Presentation style, Adding text to the Presentation. Formatting a Presentation- Adding style, Colour, gradient fills, Arranging objects, Adding Header & Footer, Slide Background, Slide layout. Adding Graphics to the Presentation- Inserting pictures, movies, tables etc into presentation, Adding Effects to the Presentation- Setting Animation & transition effect.

Reference Books:

1. Microsoft Word 2010: Level 2, by Judy Mardar and Pamela R. Toliver
2. Excel 2016 Bible 1st Edition by John Walkenbach.
3. PowerPoint For Dummies, Office 2021 Edition 1 st Edition by Doug Lowe

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Multiple Entry and Multiple Exit Option
(NEP-2020)
PART I SEM I

Title of course: IKS-I Vedic Mathematics

Course Outcomes:

After completion of this course students will be able;

1. To perform simple arithmetic calculations with speed and accuracy.
2. To generate tables of any number.
3. To perform products of large numbers quickly.
4. Apply basic Vedic techniques for fast addition, subtraction, multiplication, and division.
5. Use complements, bases, and vinculum numbers to simplify arithmetic calculations.
6. Solve quadratic and simultaneous equations and apply Vedic techniques to competitive exam problems.

UNIT I **(15 HOURS)**

Introduction to Vedas, History of Vedas History and Evolution of Vedic Mathematics Introduction of Basic Vedic Mathematics Techniques in Multiplication (Special Case, Series of 9, Series of 1 etc.), Tables etc., Various techniques to carry out basic operations covering Addition, Subtraction, Multiplication, Division, Complements and Bases, Vinculum number. Comparison of Standard Methods with Vedic Methods.

UNIT II **(15 HOURS)**

General multiplication (Vertically Cross- wise), Multiplications by numbers near base. Verifying answers by use of digital roots, Divisibility tests, Division of numbers near base, Comparison of fractions. Different methods of Squares (General method, Base method, Duplex method etc.) Cubes, Cube roots, Square Roots, General division. Quadratic Equations, Simultaneous Equations, Use of various Vedic Techniques for answering numerical aptitude questions from Competitive Examinations

Reference Books:

1. Bhatiya Dhaval, Vedic Mathematics Made Easy, Jaico Publishing House.
2. Thakur Rajesh Kumar, Vedic Mathematics for students taking Competitive Examinations. Unicorn Books 2015 or Later Edition.
3. Gupta Atul, Power of Vedic Mathematics with Trigonometry ,JaicoBooks
4. V. G. Unkalkar, Magical World of Mathematics (Vedic Mathematics), Vandana Publishers, Bangalore.

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Multiple Entry and Multiple Exit Option
(NEP-2020)
PART I SEM II

Title of course: Programming Paradigm II

Course Outcomes (COs)

By the end of this course, students will be able to:

1. Understand data structures in Python.
2. Understand concept of file handling.
3. Explore concept of GUI.
4. Use sets, tuples, and dictionaries for structured data storage and manipulation.
5. Compare list, tuple, set, and dictionary based on features and use cases.
6. Implement regular expressions for pattern matching and string processing.

UNIT I (15 HOURS)

Storing Collections of Data Using Lists: storing and accessing data in Lists, modifying Lists, operations on Lists, slicing Lists, aliasing, List Methods, Working with a List of Lists. Repeating code using loops: processing items in a List, processing characters in Strings, Looping over a range of numbers, processing Lists using Indices, reading techniques, multiline Records.

UNIT II (15 HOURS)

Storing data using other collection Types: storing data using Sets, storing data using Tuple, storing data using Dictionaries, Inverting a Dictionary. Comparison of List, Tuple, Set and Dictionary.Creating Graphical User interface: Building a Basic GUI, Models, Views, and Controllers, Customizing the Visual Style Widgets, Object-Oriented GUIs, Regular expressions.

Reference Books:

1. Practical Programming: An introduction to Computer Science Using Python, second edition, Paul ries, Jennifer Campbell, Jason Montojo, The Pragmatic Bookshelf.
2. Python for Informatics: Exploring Information, Charles Severance
3. Learning Python, Fourth Edition, Mark Lutz, O'Reilly publication

B.Sc.-M.Sc. Artificial Intelligence & Machine Learning (AI & ML)
Multiple Entry and Multiple Exit Option
(NEP-2020)
PART I SEM II

Title of course: System Analysis and Design

Course Outcomes:

After completion of this course students will be able;

1. Define and use common System Analysis and Design fundamental terminology.
2. Utilize current Analysis and Design tools to graphically characterize processes and flows in a business system.
3. Apply various fact-finding techniques for system analysis like interviews, questionnaires, and observations.
4. Conduct feasibility studies and prepare feasibility reports with technical, economic, and operational considerations.
5. Describe the role of a system analyst and differentiate between system analysis and system design.
6. Use analysis tools like DFDs, decision tables, and system development life cycle models to design effective systems.

UNIT I (15 HOURS)

Basic Concept of Systems: Definition and Concepts; Elements of a System: Input, Output Processor, Control, Feedback, Environment, Boundaries and Interface; Characteristics of a System; Types of systems -Physical and Abstract System, Open and Closed Systems, Man-made Systems; Information and its categories. System Analysis: Fact Finding Techniques: Interview - Questionnaire - Record Review - Observation. Systems Analysis: Analysing Systems Data - Feasibility Study: Technical, Economical and Operational - Steps in Feasibility Analysis, Feasibility Report, Oral Presentation

UNIT II (15 HOURS)

System Analyst - Role of a System Analyst, Introduction to System Design – Comparison between System Analysis and System Design. Analysis Tools: Data Flow Concept - Data Flow Diagram - Data Dictionary - Decision Table - Decision Tree .System Development Life Cycle -Problem Definition - Feasibility Study - Analysis - Design - Development - Implementation -Post Implementation and Maintenance. Data Flow Diagram and System Prototype Design -Meaning and Significance of Data flow diagram(DFDs)-Symbols used in DFDs -Rules for Constructing DFDs - Introduction and comparison between Physical and Logical DFDs

Reference Books:

1. S. Parthasarthy & B. W. Khalkar : System Analysis & Design, 1st Edition, Master Ed. Cons., Nashik

2. James A. Senn : Analysis & Design of Information System 2nd Edition, McGraw-Hill Int

3. V.Rajaraman – Analysis & Design of I. S. Printice Hall of India Private Ltd.2003

4. Perry Edwards: System Analysis & design Mc Graw Hill

5. System Analysis and Design Methods, Whitten, Bentaly and Barlow, Galgotia Publication.

6. System Analysis and Design Elias M. Award, Galgotia Publication

B.Sc.-M.Sc. Artificial Intelligence & Machine Learning (AI & ML)
Multiple Entry and Multiple Exit Option
(NEP-2020)
PART I SEM II

Title of course: Practical Based on Subject I DSC III

Course Outcomes:

After completion of this course students will be able to;

1. To make the student learn a programming language.
2. To learn problem solving techniques.
3. To teach the student to write programs in Python and to solve the problems.
4. Understand and demonstrate the structure and key elements of a basic Python program.
5. Use the Python interpreter and shell to perform arithmetic operations and basic scripting.
6. Apply correct indentation and syntax using identifiers, keywords, and atoms in Python code.

Lab work is based on Programming Paradigm II. This laboratory course will consist of 25 to 30. Programming exercises with focus on covering the hands-on aspects covered in theory course.

B.Sc.-M.Sc. Artificial Intelligence & Machine Learning (AI & ML)
Multiple Entry and Multiple Exit Option
(NEP-2020)
PART I SEM II

Title of course: Statistical Analysis

Course Outcomes:

After completion of this course students will be able to;

1. Acquire knowledge of concept and Meaning of bivariate data in statistics.
2. Understand the basic knowledge of Time series analysis.
3. Explain the concept and types of correlation and regression in bivariate data analysis.
4. Derive and interpret regression lines and regression coefficients from bivariate data.
5. Calculate and interpret Karl Pearson's and Spearman's correlation coefficients for ungrouped data.
6. Apply methods like progressive averages, moving averages, and least squares to identify trends and seasonal variations in time series data.

UNIT I **(15 HOURS)**

Analysis of Bivariate Data: Correlation and Regression, Concept of correlation, Types of correlation.,Methods of studying correlation: Scatter plot, Karl Pearson's correlation coefficient(r),Spearman's Rank correlation coefficient (R), Interpretation of r (with special cases r = −1, 0, and 1), Numerical problems on the computation of r and R (with and without ties) for ungrouped, data.Concept of regression, Lines of regression.Regression equations, regression coefficients, and the relation between correlation coefficients and regression coefficient. Numerical problems on ungrouped data.

UNIT II **(15 HOURS)**

Time Series, Definition and uses of time series.,Components of time series.,Methods of measuring trend: progressive averages method, moving averages method and least squares method, Seasonal variation using simple average method. Numerical problems.

Reference Books:

1. Statistical Methods, by Dr. S. P. Gupta, Sultan Chand and Sons Publication.
2. Introduction to Statistics, by C.B. Gupta.
3. Mathematical Statistics, by H.C. Saxena and J.N. Kapur.
4. Business Statistics, by S.S. Desai.

B.Sc.-M.Sc. Artificial Intelligence & Machine Learning (AI & ML)
Multiple Entry and Multiple Exit Option
(NEP-2020)
PART I SEM II

Title of course: Advanced Excel

Course Outcomes:

After completion of this course students will be able to;

1. Understand the functionalities of MS Excel.
2. Improve writing skills in different format.
3. Apply text, arithmetic, date, and time functions for data manipulation and formatting.
4. Utilize Excel features like filtering, sorting, paste special, and workbook protection.
5. Perform data analysis using What-If Analysis, data validation, logical, lookup, and array functions.
6. Create dashboards using pivot tables, slicers, and charts for visual data representation.

UNIT I **(15 HOURS)**

Introduction to Excel, Personalising Excel, Understanding and Using Basic Functions, Text Functions, Arithmetic Functions Proofing and Formatting, Protecting Excel- Excel Security , Printing Workbooks, Advance Paste Special Techniques, Time and Date Functions, Filtering and Sorting, Printing Workbooks.

UNIT II **(15 HOURS)**

What-If Analysis, Data Validation, Logical Analysis, Lookup Functions, Arrays Functions, Pivot Tables, Excel Dashboard, Slicers and Charts, VBA Macro, Introduction to VBA, Variables in VBA, Inputbox and Message Box Functions, If and select statements, Looping in VBA, Worksheet / Workbook Operations, Mail Functions – VBA

Reference Books:

1. Excel 2016 Bible, by John Walkenbach
2. Excel: Quick Start Guide from Beginner to Expert, by William Fischer
3. Mastering Advanced Excel Paperback – 21 July 2023 by Ritu Arora

B.Sc.-M.Sc. Artificial Intelligence & Machine Learning (AI & ML)
Multiple Entry and Multiple Exit
Option (NEP-2020)
PART I SEM II

Title of course: Practical Based on Subject II DSC IV

Course Outcomes:

After completion of this course students will be able to;

1. Learn the use of Arithmetic Functions in Excel
2. Learn Worksheet / Workbook Operations.
3. Perform what-if analysis, data validation, and logical operations using Excel functions.
4. Create dashboards and visualizations using pivot tables, slicers, and charts.
5. Write basic VBA macros using variables, input/output boxes, conditional and looping statements.
6. Automate tasks involving workbook operations and mailing functions using VBA.

This laboratory course will consist of 25 to 30. Excel exercises with focus on covering the hands-on aspects covered in theory course.

B.Sc.-M.Sc. Artificial Intelligence & Machine Learning (AI & ML)
Multiple Entry and Multiple Exit Option
(NEP-2020)
PART I SEM II

Title of course: Computer Architecture

Course Outcomes:

After completion of this course students will be able to;

1. To understand the structure and components of computer.
2. To familiarize a student with control unit.
3. To understand the memory subsystems.
4. Understand instruction types, addressing modes, and the instruction execution cycle.
5. Compare hardwired and microprogrammed control units, and explain datapath and control path design.
6. Analyze advanced CPU concepts such as pipelining, RISC vs CISC, and superscalar architecture.

UNIT 1 **(15 HOURS)**

Function and structure of a computer, Functional components of a computer, Interconnection of components, Performance of a computer, Introduction to Computer Organization, CPU Organization Memory subsystem Organization, and Interfacing, I/O Subsystem Organization and Interfacing, a relative Simple Computer, Software, hardware interaction, layers in computer architecture, Central processing and machine language , instructions, addressing modes, instruction types, instruction set selection, Instruction and execution cycle.

UNIT 2 **(15 HOURS)**

Control Unit: Data path and control path design, microprogramming v/s hardwired control, Pipelining in CPU designing, RISC v/s CISC, Superscalar processors. Memory Subsystems: Storage technologies, memory array organization, memory hierarchy, interleaving, cache memory, Auxiliary memory, Associative Memory and virtual memory.

Reference Books:

1. C. Hamacher, Z. Vranesic and S. Zaky, "Computer Organization", McGrawHill,
2. W. Stallings, "Computer Organization and Architecture - Designing for Performance", Prentice Hall of India,
3. Computer System Architecture, M. Morris Mano, Third Edition, Pearson Education
4. D. A. Patterson and J. L. Hennessy, "Computer Organization and Design - The Hardware/Software Interface", Morgan Kaufmann.

B.Sc.-M.Sc. Artificial Intelligence & Machine Learning (AI & ML)
Multiple Entry and Multiple Exit Option
(NEP-2020)
PART I SEM II

Title of course: Computer Network

Course Outcomes:

After completion of this course students will be able to;

1. Understand with switching and routing concepts in networking technologies.
2. This course will help a student understand the various network protocols.
3. Familiar with IPV4 and IPV6 address.
4. Explain link-layer addressing, DLC services, and protocols like HDLC, PPP, and MAC techniques.
5. Describe the functioning of wired and wireless LAN technologies including IEEE 802.11 and Bluetooth.
6. Apply unicast and multicast routing algorithms to determine optimal data transmission paths.

UNIT I **(15 HOURS)**

Introduction, Link-Layer Addressing, DLC Services, Data-Link Layer Protocols, HDLC, PPP, Media Access Control, Wired LANs: Ethernet, Wireless LANs, Introduction, IEEE 802.11, Bluetooth, Connecting Devices.Network Layer Services, Packet switching, Performance, IPV4 Addresses, Forwarding of IP Packets, Network Layer Protocols: IP, ICMP v4, Unicast Routing Algorithms, Protocols, Multicasting Basics, IPV6 Addressing,

IPV6 Protocol.

UNIT II **(15 HOURS)**

Transport Layer, Process to process delivery, TCP-UDP, Operation and uses, Three-way Handshake, for connection establishment and termination. Application Layer, Domain Name Space, Remote Logging, Electronic Mail - File Transfer- Email, FTP.WWW and HTTP-HTTP.

Reference Books:

1. Black C “Computer networks protocols, standards and Interface”, prentice hall of India,
2. Stlling W, “Computer communication network” (4th Edition), prentice hall of India, 1993
3. Tanenbaum A.S. “Computer Network”, prentice hall of India, 1981
4. Forouzan, “TCP/IP Protocol Suite”, Tata McGraw Hill.
5. Walrand &Varaiya, “High Performance Communication Networks”, 2/e, Elsevier”, 2003

B.Sc.-M.Sc. Artificial Intelligence & Machine Learning (AI & ML)
Multiple Entry and Multiple Exit Option
(NEP-2020)
PART I SEM II

Title of course: Practical based on Subject III DSC III and Subject III DSC IV

Course Outcomes:

After completion of this course students will be able to;

1. The students will understand the function of all components of Computer architecture.
2. The students will learn how to design microprogramming and hardwired control unit.
3. Understand different advanced network technologies that can be used to connect different networks.
4. Understand different network technologies and their application.
5. Configure and compare wired and wireless LANs using tools like Cisco Packet Tracer or Wireshark.
6. Use simulation or browser-based tools to demonstrate application layer protocols such as DNS, FTP, email, and HTTP.

Lab work is based on Computer Architecture and Basics of Networking. This laboratory course will consist of 25 to 30. Programming exercises with focus on covering the hands-on aspects covered in theory course.

B.Sc.-M.Sc. Artificial Intelligence & Machine Learning (AI & ML)
Multiple Entry and Multiple Exit Option
(NEP-2020)
PART I SEM II

Title of course: OE-II: Basics of HTML

Course Outcomes:

After completion of this course students will be able to;

1. Understand concepts of Tags in HTML.
2. Understand basics of CSS.
3. Design web pages using forms, iFrames, anchor tags, and structural elements like div and span.
4. Use various CSS types, selectors, and properties to style text, background, fonts, and layout.
5. Implement layout and positioning techniques using the CSS box model, float, and display properties.
6. Apply basic HTML tags to format text, create lists, tables, hyperlinks, and embed multimedia.

UNIT I **(15 HOURS)**

HTML: Introduction to web programming, Introduction to HTML, History of HTML, Editors, Basic Formatting Tags,-Grouping Using Div Span, Lists, Working with Hyperlinks, Images and Multimedia, Tables, iFrame, Forms, Headers, Anchor tags & hrefs

UNIT II **(15 HOURS)**

CSS: Introduction, Benefits of CSS, Types of CSS, Selectors, Color Background, Cursor, Text, Fonts, Span, opacity, List Tables, Box Model, Display Positioning, Floats, Dividing up content with the div tag, CSS borders

References:

1. Complete HTML- Thomas Powel HTML Black Book- Steven Holzner
2. Teach Yourself Web Technologies – Ivan Bayross
3. HTML and CSS 3- Seventh edition – Castro Elizabeth and Bruce Hyslop.

B.Sc.-M.Sc. Artificial Intelligence & Machine Learning (AI & ML)
Multiple Entry and Multiple Exit
Option (NEP-2020)
PART I SEM II

Title of course: VEC- I Democracy, Election and Constitution

Syllabus will be provided by Shivaji University as per NEP

Course Outcomes:

After completion of this course students will be able to;

1. The overall course of political science develops the understanding of the basic concepts of Politics, Political theory and Political science.
2. Social democracy thus aims to create the conditions for capitalism to lead to greater democratic, egalitarian and solidaristic outcomes
3. A course focused on democracy, elections, and the constitution typically aims to equip students with a strong understanding of democratic principles.
4. Understand the concept, principles, and types of democracy, including direct and representative democracy.
5. Explain the role of decentralization and the 73rd and 74th Constitutional Amendments in grassroots governance.
6. Evaluate the structure and process of elections at national, state, and local levels, and the role of the State Election Commission.